Notes: Einav, Finkelstein, and Cullen (QJE, 2010)
Estimating Welfare in Insurance Markets Using Variation in Prices

Contents

1 Big picture

- **Question.** How can researchers estimate the welfare loss from inefficient pricing in insurance markets with selection using relatively transparent empirical objects?
- **Core idea.** In insurance markets, price affects not only the quantity of insurance purchased but also the composition of buyers. Because the set of insured consumers changes with price, insurers' average costs and marginal costs are endogenous to price.
- **Sufficient-statistics approach.** The paper argues that the demand curve and the cost curve are enough to evaluate welfare consequences of pricing existing insurance contracts. The researcher does not need to fully specify consumers' risk preferences, private information, or utility functions.
- **Graphical contribution.** The paper gives a simple supply-and-demand style representation of welfare loss under adverse selection and advantageous selection. The same logic used in intermediate microeconomics can be applied to insurance selection once the cost curve is allowed to move with the pool of buyers.
- **Empirical strategy.** Identifying price variation estimates demand. The same price variation, combined with claims or cost data, estimates how costs vary as the insured pool changes. The slope of the cost curve directly tests the existence and direction of selection.
- **Application.** The empirical example uses employer-provided health insurance at Alcoa in 2004. The key sample contains 3,779 salaried employees with family coverage who chose between two PPO options.
- **Source of variation.** Alcoa's business-unit structure created different relative prices for the same high- and low-coverage options. The authors argue that for salaried employees this price variation is plausibly unrelated to demand and cost determinants.
- **Main empirical result.** The application finds adverse selection: when the high-coverage option is more expensive, the employees who still choose it have higher average incremental costs.
- **Quantitative welfare result.** Despite detecting adverse selection, the estimated welfare loss from inefficient competitive pricing is small in this setting: about $9.55 per employee per year, roughly 3% of the total surplus from efficient pricing.
- **Policy implication.** Detecting adverse selection is not enough. The welfare cost may be small, and common policy tools such as subsidies or mandates may cost more than the inefficiency they are designed to correct.

2 Incremental contribution of the paper

- **From detection to quantification.** Much of the insurance literature tests whether asymmetric information exists. This paper asks how large the welfare loss is once selection is detected.
- **A transparent welfare framework.** Rather than building a full structural model of primitives, the paper shows that demand and cost curves are sufficient statistics for welfare analysis of pricing existing contracts.

- **A unified graphical tool.** The framework visually explains why adverse selection creates underinsurance and advantageous selection creates overinsurance under average-cost pricing.
- **A direct test of selection.** The slope of the cost curve provides a direct test of selection. If marginal cost changes with price-induced selection into insurance, then buyers differ systematically in expected costs.
- **Separation from moral hazard.** The cost-curve test is not mechanically confounded by moral hazard because it compares the costs of people who all chose the same contract, while the sample changes because of price.
- **Practical data contribution.** The approach requires only prices, quantities, and realized costs or claims. These data are often more available in insurance markets than cost data are in standard product markets.
- **Policy relevance.** The method can evaluate subsidies, mandates, price regulation, and other interventions that change the prices of existing contracts.
- **Limit of the contribution.** The approach does not estimate welfare effects from changing the set of insurance contracts offered. For new contract design or contract-space distortions, a richer model of primitives is still needed.
- **Empirical illustration.** The Alcoa application shows that adverse selection can exist while producing a modest welfare loss, which cautions against equating market failure detection with large policy gains.

3 Data and measurement

3.1 Insurance setting and sample

- The empirical application studies U.S.-based employees of Alcoa, Inc. in 2004.
- Alcoa introduced a new set of health insurance options to nearly all salaried employees and about half of hourly employees.
- The authors focus on salaried employees with family coverage who chose one of the two modal PPO options.
- The final baseline sample contains 3,779 employees.
- The two key contracts are a higher-coverage option, denoted H, and a lower-coverage option, denoted L.

Interpretation: The sample is deliberately narrow. Restricting to salaried workers and family coverage makes the price variation cleaner and the two-contract framework closer to the theory.

3.2 Price variation

- The empirical price is the relative annual premium for high coverage: $p_i = p_i^H - p_i^L$.
- Alcoa headquarters offered a fixed menu of coverage options but allowed business-unit presidents to choose among price menus.

- The same contracts could therefore have different relative prices across otherwise similar salaried employees.
- In the baseline sample, relative prices range from \$384 to \$659.
- The paper emphasizes that this was the first year in which business-unit presidents made these benefit-pricing decisions, which reduces concerns that pricing was finely optimized using employee risk information.

Interpretation: The design needs prices to shift demand but not directly reflect unobserved health risk or willingness to pay. The business-unit pricing structure is the key quasi-experimental source of variation.

3.3 Demand measure

- The demand outcome is a binary indicator D_i equal to one if employee i chooses contract H and zero if employee i chooses contract L.
- Demand is estimated as a function of the relative price p_i.
- In the raw data, the fraction choosing high coverage generally falls as the relative price rises.
- In the baseline linear specification, a \$100 increase in the relative price of high coverage lowers the probability of choosing high coverage by about seven percentage points.

Interpretation: Demand gives willingness-to-pay information. The downward-sloping demand estimate is the first input into welfare analysis.

3.4 Cost measure

- The relevant cost is not total medical spending. It is the incremental insurer cost of covering the same realized medical claims under contract H rather than contract L.
- Let m_i denote employee i's realized family medical expenditures. The cost variable is

$$c_i = c(m_i; H) - c(m_i; L).$$

- The authors compute this cost using detailed claims and plan rules.
- Contract H and contract L mainly differ in deductibles and out-of-pocket maximums.
- Because the contracts are otherwise similar, the incremental-cost calculation isolates the insurer's additional cost of higher coverage.

Interpretation: Cost data are the second key input. The central empirical question is whether average incremental cost among high-coverage buyers changes as price changes.

3.5 Exogeneity and balance checks

- Table I compares salaried employees who faced the lowest relative price with employees who faced higher relative prices.

- Observable characteristics are broadly balanced: age, gender, race, salary, spouse age, family size, child age, and prior medical spending are not strongly different across price groups.
- The two possible weak exceptions are tenure and prior medical spending, but the paper reports that a joint test cannot reject that the covariates are jointly unrelated to price.
- The analogous exercise for hourly employees looks less balanced, which supports the decision to focus on salaried employees.

Interpretation: Balance is essential because the same price variation identifies both demand and cost. If price were set based on expected costs, the cost curve estimate would not cleanly measure selection.

4 Models and empirical specifications

4.1 Theoretical demand and cost curves

Each individual has characteristics ζ_i, a willingness to pay for high coverage $\pi(\zeta_i)$, and expected cost $c(\zeta_i)$. Aggregate demand for insurance is

$$D(p) = \Pr\left(\pi(\zeta_i) \geq p\right). \tag{1}$$

Average cost is the expected cost of those who choose high coverage at price p:

$$AC(p) = E\left[c(\zeta_i) \mid \pi(\zeta_i) \geq p\right]. \tag{2}$$

Marginal cost is the expected cost of the marginal buyer:

$$MC(p) = E\left[c(\zeta_i) \mid \pi(\zeta_i) = p\right]. \tag{3}$$

Interpretation: Selection is embedded in the cost curves. If the marginal buyers entering as price falls are cheaper or more expensive than existing buyers, average and marginal costs move with price.

4.2 Adverse versus advantageous selection

- **Adverse selection.** Consumers with the highest willingness to pay also have the highest expected costs. The marginal cost curve slopes downward in quantity. Average-cost pricing leads to too little high coverage.
- **Advantageous selection.** Consumers with the highest willingness to pay have the lowest expected costs. The marginal cost curve slopes upward in quantity. Average-cost pricing leads to too much high coverage.
- In both cases, the competitive equilibrium is determined by average-cost pricing, whereas the efficient allocation is determined by comparing willingness to pay with marginal cost.

Interpretation: The inefficiency comes from the fact that firms must charge a uniform price even though buyers differ in expected cost.

4.3 Baseline demand and cost estimating equations

The empirical application estimates linear demand and cost equations:

$$D_i = \alpha + \beta p_i + \varepsilon_i, \tag{4}$$

$$c_i = \gamma + \delta p_i + u_i. \tag{5}$$

- D_i equals one if employee i chooses high coverage.
- p_i is the relative annual premium of contract H over contract L.
- c_i is the incremental insurer cost of providing contract H instead of contract L.
- The demand equation is estimated on all employees in the baseline sample.
- The average-cost equation is estimated on the endogenously selected sample of employees who choose contract H.

Interpretation: The cost regression asks: when price is higher and only people with higher willingness to pay remain in high coverage, are those people more costly?

4.4 Recovering marginal cost and welfare

Given the linear demand and average-cost equations, the marginal cost curve is recovered from total cost:

$$MC(p) = \frac{\partial(AC(p)D(p))}{\partial D(p)} = \frac{1}{\beta}\frac{\partial[(\alpha+\beta p)(\gamma+\delta p)]}{\partial p} = \frac{\alpha\delta}{\beta} + \gamma + 2\delta p. \tag{6}$$

- The competitive outcome satisfies $p = AC(p)$.
- The efficient outcome satisfies $p = MC(p)$.
- The welfare loss from adverse selection is the triangle between demand and marginal cost over the quantity range that is efficient but not purchased in equilibrium.
- In the paper's linear specification, the welfare-loss triangle is computed using the estimated demand, average-cost, and marginal-cost curves.

Interpretation: Once demand and cost curves are estimated, welfare calculations are mechanical and visually transparent.

4.5 Policy counterfactuals

- The framework can evaluate pricing policies for existing contracts: subsidies, mandates, price regulation, and restrictions on allowable pricing differences.
- In the Alcoa application, a subsidy large enough to move the market to the efficient allocation would have a social cost of about \$45 per employee using a marginal cost of public funds of 0.3.

- This policy cost is much larger than the estimated welfare gain of $9.55 per employee.
- The authors also estimate that monopoly pricing would create a much larger welfare loss, just below $100 per employee.

Interpretation: The welfare framework allows the paper to compare the size of the selection inefficiency to the costs and benefits of plausible policy responses.

5 Identification and empirical design

5.1 What the design identifies

- The paper identifies the demand curve and the endogenous cost curve for existing insurance contracts using cross-sectional price variation.
- The demand curve identifies willingness to pay for contract H relative to contract L.
- The cost curve identifies how expected insurer costs vary as the composition of high-coverage buyers changes with price.
- The slope of the cost curve provides a direct test of selection.
- The combination of demand and cost identifies welfare from alternative prices for the same contracts.

Interpretation: The design does not require observing individual private information. It only requires observing how choices and costs respond to plausibly exogenous price changes.

5.2 Why price variation is central

- Without price variation, one might compare high-coverage and low-coverage buyers and detect a correlation between coverage and costs.
- That comparison cannot distinguish selection from moral hazard because people with more coverage may spend more because coverage changes behavior.
- Price variation changes who selects into high coverage while holding the observed contract fixed for the cost-curve sample.
- Therefore, the cost-curve slope measures how risk composition changes with price-induced selection.

Interpretation: The empirical advantage is not simply having claims data; it is having claims data plus variation in the price of the same insurance options.

5.3 Selection versus moral hazard

- The cost-curve test is robust to the existence of moral hazard because the cost equation is estimated among people choosing contract H.

- As price changes, the sample of high-coverage buyers changes, but everyone in that sample has the same coverage.
- The authors discuss how moral hazard could be estimated separately by comparing cost curves under H and L if the researcher has the necessary data.
- In the Alcoa application, the authors do not find evidence that moral hazard is driving the welfare estimate.

Interpretation: The paper's main selection test is conceptually distinct from the usual positive-correlation test, which jointly captures selection and moral hazard.

5.4 Limits of the design

- The approach analyzes pricing of existing contracts, not welfare losses from missing or distorted contracts.
- It relies on uncompensated demand, so it does not fully model income effects.
- Functional-form assumptions matter when the observed price variation does not span the equilibrium and efficient prices.
- The Alcoa application is one firm, one year, and a specific set of health insurance choices. Its quantitative welfare estimate should not be generalized mechanically to all health insurance markets.

Interpretation: The method is deliberately transparent and portable, but it trades off richness in counterfactual contract design for fewer assumptions about underlying primitives.

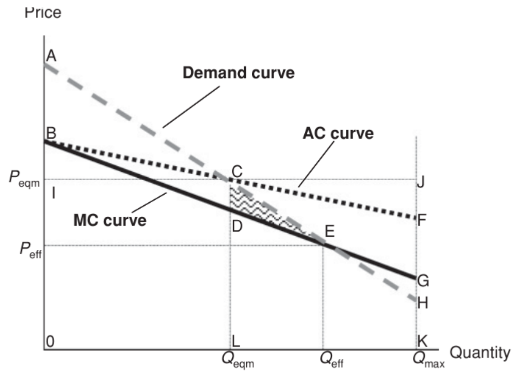
6 Tables and figures

6.1 Figures I and II: Welfare loss under adverse and advantageous selection

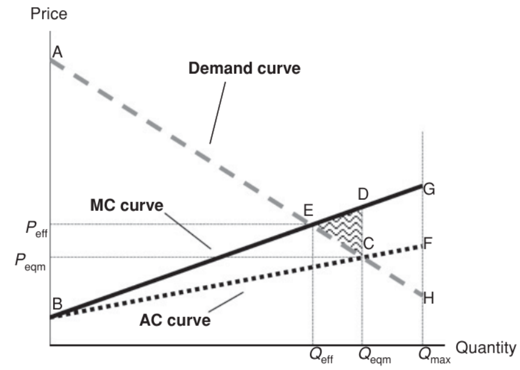
Read:

- Figure I shows adverse selection. High-willingness-to-pay consumers have high expected costs, so marginal cost falls as quantity expands. Average-cost pricing produces underinsurance.
- In Figure I, the competitive equilibrium is where demand intersects average cost, while the efficient allocation is where demand intersects marginal cost.
- The shaded triangle is the welfare loss from people who would value high coverage above marginal cost but do not buy it at the average-cost price.
- Figure II shows advantageous selection. High-willingness-to-pay consumers have low expected costs, so marginal cost rises as quantity expands. Average-cost pricing produces overinsurance.
- The two graphs clarify why selection does not always imply too little insurance; the sign of selection determines the direction of the distortion.

Economic message: Welfare analysis in insurance markets requires both demand and cost curves. The sign and magnitude of selection are empirical questions.



(a) Figure I: adverse selection and underinsurance



(b) Figure II: advantageous selection and overinsurance

6.2 Table I and Table II: Price variation and raw demand-cost patterns

Read:

- Table I checks whether employees facing the lowest relative price differ from employees facing higher prices.
- Most observables are similar across price groups: age, gender, race, salary, spouse age, family size, and age of youngest child.
- Prior medical spending is somewhat lower for employees facing higher prices, but the paper reports that the observables are not jointly predictive of price.
- Table II shows the raw relationship between relative price, high-coverage choice, and average incremental cost.
- Demand is downward sloping: the fraction choosing high coverage is generally lower when the relative price is higher.
- Average incremental costs among high-coverage buyers tend to be higher at higher prices, foreshadowing the adverse-selection result.

Economic message: Table I supports the price-exogeneity assumption; Table II displays the two empirical ingredients of the paper: demand falls with price and costs rise among the selected insured pool.

TABLE I
ASSESSING THE EXOGENEITY OF THE PRICE VARIATION

	Faced lowest relative price (2,939 employees) (1)	Faced higher relative prices (840 employees) (2)	Difference (3)	Coefficient (4)	p-value (5)
Age (mean)	42.74	42.40	0.33	−0.245	.31
Tenure (mean)	13.02	11.63	1.39	−0.565	.08
Fraction male	0.862	0.852	0.009	1.268	.79
Fraction white	0.874	0.825	0.049	−6.998	.40
Log(annual salary) (mean)	11.16	11.05	0.11	−8.612	.17
Spouse age (mean)	41.37	41.05	0.32	−0.200	.41
Number of covered family members (mean)	4.14	4.07	0.07	−1.400	.36
Age of youngest covered child (mean)	9.81	9.41	0.40	−0.3	.26
2003 medical spending (in US$)[a]					
All (mean)	7,027	5,922	1,105	−0.0001	.09
In most common 2003 plan (mean)	6,938	5,967	971	−0.0001	.10

(a) Table I: balance and exogeneity of price variation

TABLE II
THE EFFECT OF PRICE ON DEMAND AND COSTS

(Relative) price ($) (1)	Number of employees (2)	Fraction chose contract H (3)	Average incremental cost ($) for those covered under: Contract H (4)	Contract L (5)
384	2,939	0.67	451.40	425.48
466	67	0.66	499.32	423.30
489	7	0.43	661.27	517.00
495	526	0.64	458.60	421.42
570	199	0.46	492.59	438.83
659	41	0.49	489.05	448.50

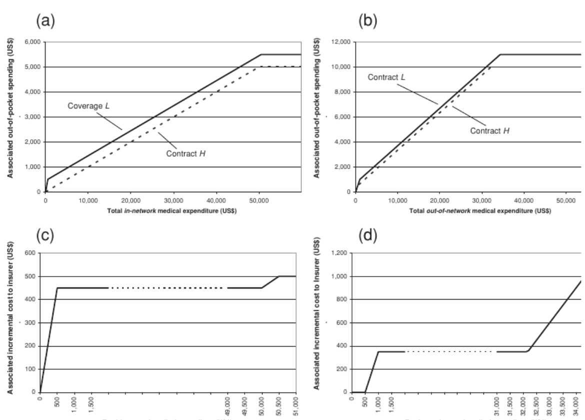
(b) Table II: price, demand, and costs

6.3 Figures III and IV: Contract rules and incremental insurer costs

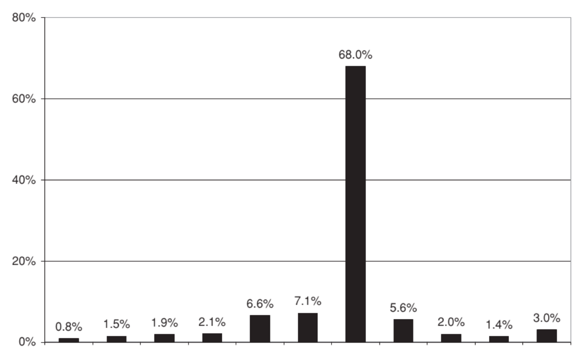
Read:

- Figure III describes the cost-sharing rules of contracts H and L for in-network and out-of-network spending.
- The top panels show out-of-pocket spending under the two contracts; contract H has lower consumer cost sharing.
- The bottom panels translate those contract differences into incremental insurer costs from providing H rather than L.
- Figure IV shows the distribution of incremental insurer costs c_i in the baseline sample.
- A large mass appears at $450 because many employees have medical spending in the range where the contract difference mechanically implies that incremental cost.

Economic message: The cost variable is not a black-box claims measure. It is mechanically constructed from plan rules and realized claims, which makes the cost-curve interpretation clean.



(a) Figure III: contract rules and incremental-cost construction



(b) Figure IV: distribution of incremental insurer costs

6.4 Table III and Figure V: Baseline estimation and empirical welfare triangle

Read:

- Table III reports the baseline linear estimates. The demand coefficient on price is -0.00070, so a $100 price increase reduces high-coverage take-up by about seven percentage points.
- The average-cost coefficient on price is 0.15524, meaning the average incremental cost of high-coverage buyers rises as the price rises.
- The positive cost slope implies adverse selection: the people who remain in high coverage at higher prices are more costly.
- The implied competitive outcome is $Q = 0.617$ and $P = 463.5$.
- The implied efficient outcome is $Q = 0.756$ and $P = 263.9$.

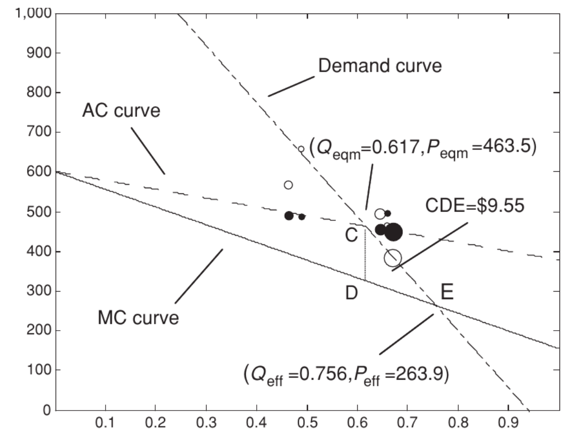
- The estimated welfare cost from selection is \$9.55 per employee, while total surplus from efficient allocation is \$283.39.
- Figure V maps the estimates into the same welfare diagram as Figure I and shows the empirical triangle CDE.

Economic message: The paper detects adverse selection, but the welfare loss from the resulting pricing inefficiency is modest in this application.

TABLE III
ESTIMATION RESULTS

Dependent variable (sample)	1 if chose High (both High and Low) (1)	Incremental cost (only High) (2)
Panel A: Estimation results		
Relative price of High (US\$)	−0.00070	0.15524
	(0.00032)	(0.06388)
	[.034]	[.021]
Constant	0.940	391.690
	(0.123)	(26.789)
	[.000]	[.000]
Mean dependent variable	0.652	455.341
Number of observations	3,779	2,465
R^2	.008	.005
Panel B: Implied quantities of interest		
Competitive outcome (point C in Figure I)		$Q = 0.617$, $P = 463.5$
Efficient outcome (point E in Figure I)		$Q = 0.756$, $P = 263.9$
Efficiency cost from selection (triangle CDE)		9.55
Total surplus from efficient allocation (triangle ABE)		283.39
Efficiency cost from mandating contract H (triangle EGH)		29.46

(a) Table III: baseline demand, cost, and welfare estimates



(b) Figure V: empirical welfare triangle

7 Short summary

- The paper develops a transparent framework for welfare analysis in insurance markets with selection.
- The core empirical objects are the demand curve and the cost curve.
- Price variation identifies demand and, with claims data, identifies how insurer costs vary as the insured pool changes.
- A non-flat cost curve is direct evidence of selection; the sign of the slope indicates whether selection is adverse or advantageous.
- The framework does not require a full structural model of utility, risk preferences, or private information.
- It can evaluate policies that change the prices of existing contracts, including subsidies, mandates, and price regulation.
- The Alcoa application uses 2004 employer health insurance choices and claims for salaried employees with family coverage.
- The empirical estimates show adverse selection: high-coverage buyers are costlier when price is higher.

- The estimated welfare loss from inefficient competitive pricing is about $9.55 per employee per year.
- The main lesson is that market failures should be quantified, not merely detected.

8 Conclusion

8.1 Core conclusion

Einav, Finkelstein, and Cullen show that welfare analysis in insurance markets with selection can be done using demand and cost curves. The approach turns a difficult hidden-information problem into a familiar pricing-and-quantity problem: estimate how many people buy insurance at different prices, estimate how costly those selected buyers are, and compare willingness to pay with marginal cost.

8.2 Economic intuition

The intuition is that adverse selection changes the supply side of the market. When price is high, only consumers with high willingness to pay remain insured. If those consumers are also high cost, average costs rise with price and average-cost pricing can exclude consumers whom it would be efficient to insure. The welfare loss is not the mere existence of high-cost buyers; it is the gap between willingness to pay and marginal cost for people who are inefficiently priced out of coverage.

8.3 Incremental contribution revisited

The paper's contribution is methodological and empirical. Methodologically, it gives a sufficient-statistics framework for selection markets that is far easier to implement and compare across settings than a fully specified structural model. Empirically, it shows that even when adverse selection is present, the welfare loss from inefficient pricing can be small. This result reframes the empirical question from "Is there asymmetric information?" to "How much welfare does the resulting selection actually destroy?"

8.4 Implications for policy

The policy implication is cautious. Adverse selection provides a theoretical rationale for government intervention, but the optimal intervention depends on magnitudes. In the Alcoa application, the estimated social cost of a subsidy sufficient to reach the efficient allocation is much larger than the welfare gain. Mandatory high coverage also appears more costly than competitive pricing in the authors' estimates. Thus, detecting adverse selection does not automatically justify aggressive intervention.

8.5 Limitations

The framework only studies pricing of existing contracts. It does not capture welfare losses from missing contracts, endogenous contract design, or the introduction of new coverage options. The empirical application is limited to one employer, one year, and a specific sample of salaried employees with family coverage. The welfare estimates therefore should not be treated as universal estimates for health insurance markets.

8.6 Takeaway

The enduring takeaway is that insurance-market welfare analysis can be made transparent. With credible price variation and cost data, researchers can estimate demand, estimate endogenous costs, test for the direction of selection, and compute welfare triangles. The paper's broader message is that market failures matter economically only to the extent that their welfare costs are quantitatively meaningful.